

INDIA'S INCENTIVES

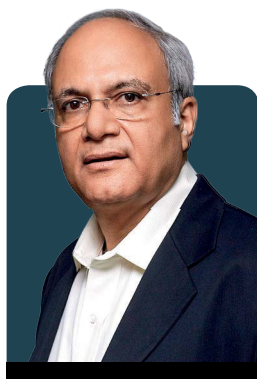


It is around 35% of project cost globally



In India, between the centre and states, it is around 70% of the project cost (50% by the centre, and an additional 20% by the states)

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“Globally, there are many fabs coming up which will not have ATMPs sitting next to them. So, there is an opportunity in packaging... Given that most of the packaging market is in Asia, that is where India can benefit”

AJAI CHOWDHRY

FOUNDER, ELECTRONICS PRODUCTS INNOVATION CONSORTIUM; CO-FOUNDER, HCL

ning around OSATs,” says Faruqui. He says China understands the strategic importance of OSATs.

China has three OSAT players in the next-generation advanced packaging, 11 in the commercial or current age advanced packaging and five in the legacy playing architectures—a portfolio unmatched in any geography. Fab Economics says China-based OSAT facilities took in annual revenue of \$14.68 billion in 2023, which could grow four times by the decade’s end.

US-China tensions have been going up in the past few years. The US has imposed targeted sanctions on advanced technology across equipment and fabrication services to be exported to Chinese players, including restricting TSMC wafer supply

used for leading-edge products.

However, according to the Global Business Policy Council R&A, advanced packaging is still untouched. Instead, Taiwan-based OSATs with manufacturing sites in China, such as King Yuan Electronics and ASE Group, have aggressively ramped up their capacity support for China-based clients, including HiSilicon, Oppo, and Unisoc.

Faruqui notes that China has been developing OSAT capacity aggressively for the past two years as it knows that the end value chain would not be a permanent free pass.

“India is a very close alternative to China for an OSAT hub, but an India-based OSAT ecosystem build-up requires strategic planning informed with China OSAT

portfolio,” says Faruqui, adding, “Otherwise, OSAT being a cost-sensitive play, the Indian player may face trouble commercially.”

NOT JUST CHINA

Indian firms will again have to get a technology-grade licence. The IESA’s Gupta says the production-grade technology for OSAT can be obtained from various sources and the international players that own the IP and have set up plants in China, Taiwan, Malaysia and Singapore. For instance, CG Power has signed a JV with Renesas Electronics and Stars Microelectronics. “This relationship covers both access to technology through Stars Microelectronics and offtake of output by Renesas Electronics, which are two most important ingredients for commercial success of an OSAT project,” says Satya Gupta, President of VLSI Society.

But semiconductor analyst Arun Mampazhy warns of the danger of expanding in plain-vanilla packaging and too many companies setting up facilities, which could lead to idle capacity. To avoid this pitfall, the newcomers could graduate to advanced packaging.

Advanced packaging architectures lie behind all semiconductor leadership products that are the true moat of competitive advantage of end devices. This kind of packaging architecture is behind Nvidia’s H100 GPUs powering Gen AI/ChatGPT. The same goes for AMD MI 300 AI Accelerators, Apple M series Ultra, A17, Intel’s Meteor Lake and Amazon’s Inferno.

Prithvideep Singh, General Manager at CDIL Semiconductors, a Mohali-headquartered firm that began making chips in 1964 but now works on a “fabless” model, says these OSATs can look at developing products five years down the line. “It might take five years, but step